Rethinking case attraction on Ancient Greek infinitive clauses

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(1) a. συμβουλεύει **τῷ Ξενοφῶντι ἐλθόντα** εἰς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
Ele aconselha Xenofonte ir a Delfos e interrogar o deus sobre a viage. (Xen. *Anab.* 3.1.5)

b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τἀληθῆ.
Ele me permitiu vir frente a vós dizer a verdade. (Xen. *Hell.* 6.1.13)

Trabalhos mencionados: Lancelot (1658, 1709), Lakoff (1970), Andrews (1971), Quicoli (1982), Spyropoulos (2005), Tantalou (2003), Sevdali (2007, 2013)

Fonte dos dados: Diorisis Corpus (Vatri e McGillivray 2018, 2020) with help of the project's querying tool, Diorisis Search (Vatri 2020-2022).

***Corpus*:** Sentenças coletadas a partir das fontes literárias áticas e Heródoto.

Ferramentas: R (R Core Team 2013) com o os pacotes tidyverse (Wickham et al. 2019), dagitty (Textor et al. 2016), ggdag (Barrett 2025) e rethinking (McElreath 2020), bem como o software Stan (Stan Development Team 2025a,b).

Modelo tipológico: Corbett (2006)

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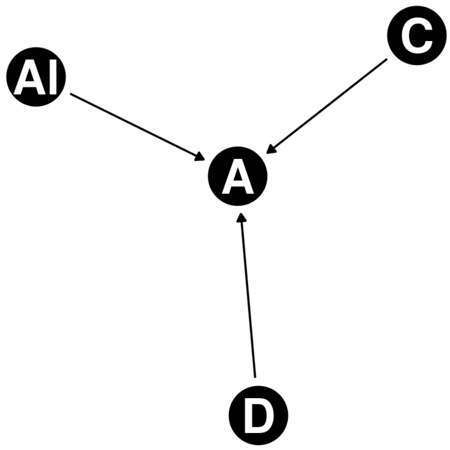


Infinitivos copulares: ver Kühner e Gerth (1898)

(2) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):

→ Noun
Adjective
↗
Participle
↗
Attributive → Predicative → Relative Pronoun → Personal Pronoun
Verb
- ⟷ +
Likelihood of agreement with semantic justification

Verbos impessoais: ver Humbert (1945)

Verbos com semântica modal: verbos como ἐξαρκεῖ e ἔξεστί tem maior probabilidade de apresentar atração de caso (Geraldes 2020, 2025).

(3) ἀλλ’ ἐξαρκῇ σοι τὰ ἔργα αὐτῶν ὁρῶντι σέβεσθαι καὶ τιμᾶν τοὺς θεούς.
Mas é suficiente a ti observar as ações e respeitar e honrar os deuses. (Xen. *Mem.* 4.3.13)

Autoria e dialeto: Geraldes (2020, 2025) mostra que o uso de verbos com semântica modal são mais frequentes em diálogos filosóficos que textos historiográficos.

Ordem de palavra e distância: maiores distâncias diminuem a probabilidade de atração de caso. Alvos que antecedem o controlador parecem exigir atração de caso como no exemplo (4).

(4) ὡς **ἀγαθοῖς** τε **ὑμῖν** προσήκει εἶναι.
Que vos seja dado ser boa gente. (Xen. *Anab.* 3.2.11)

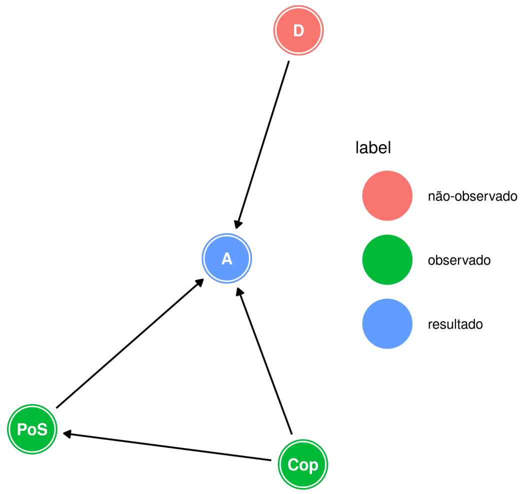


Figura 1: DAG with **A**ttraction, **D**omain, **P**art **o**f **S**preech and **Cop**ula

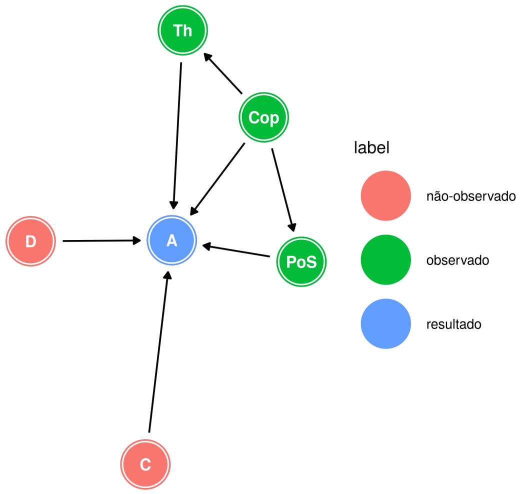


Figura 2: DAG with **A**ttraction, **D**omain, **C**ontroller **P**art **o**f **S**preech, **Cop**ula and **Th**ematic role.

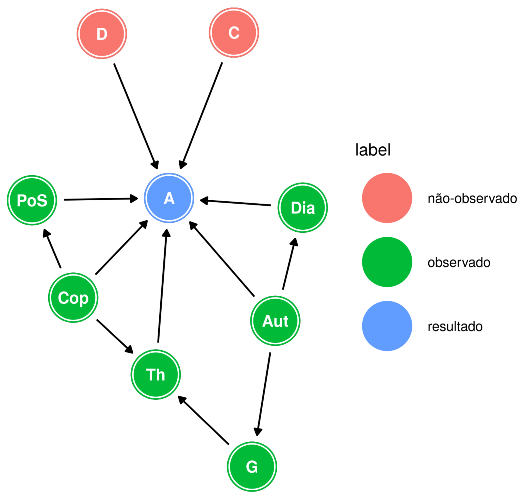


Figura 3: DAG with **A**ttraction, **D**omain, **C**ontroller **P**art **o**f **S**preech, **Cop**ula, **Th**ematic role, **Aut**horship, **G**enre, and **Dia**lect.

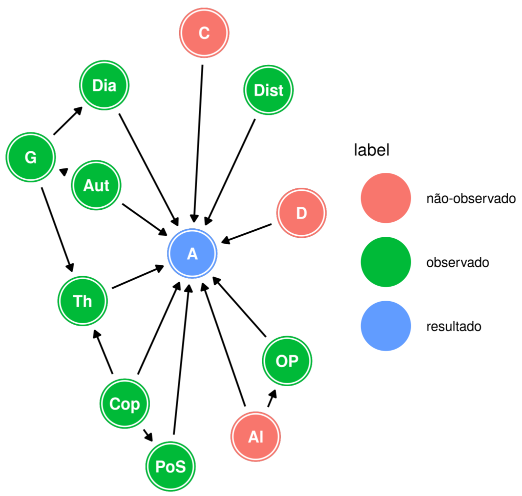
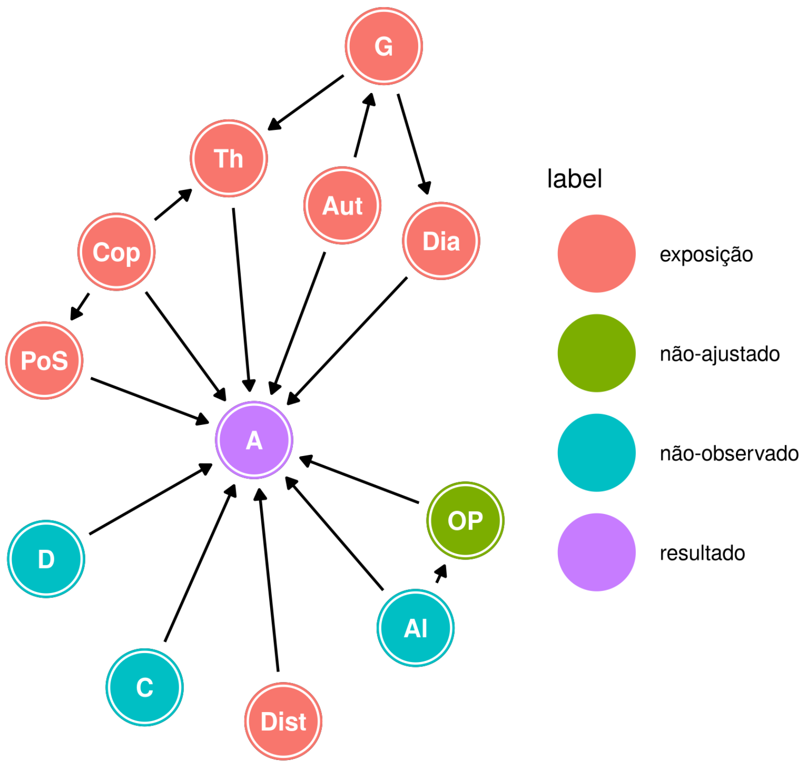


Figura 4: DAG with **A**ttraction, **D**omain, **C**ontroller, **T**arget, **P**art **o**f **S**preech, **Cop**ula, **Th**ematic role, **Aut**horship, **G**enre, **Dia**lect, **Dist**ance, and **W**ord **O**rder.

Modelo causal final:



Modelo linear generalizado:

$$
\begin{aligned}
\text{A} &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha_{\text{Sent}} + \beta_{\text{Cop}}\text{Cop}_i + \beta_{\text{Dist}}|\text{Dist}_i| + \\
&+ \beta_{\text{PoS}} \sum_{j=0}^{\text{PoS}_i-1} \delta_{j_{\text{PoS}}} \\
&+ \beta_{\text{Th}} \sum_{j=0}^{Th_i-1} \delta_{j_{\text{Th}}} \\
&+ \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i} \\
\alpha &= \overline{\alpha}_{\text{Sent}} + z_{\text{Sent}} * \sigma_{\text{Sent}} \\
\beta_{\text{Cop}}, \beta_{\text{Dist}} &\sim \text{Normal}(0, 1) \\
\beta_{\text{PoS}}, \beta_{\text{Th}} &\sim \text{Normal}(0, 1) \\
\delta_{\text{PoS}}, \delta_{\text{Th}} &\sim \text{Dirichlet}(2) \\
\beta_{\text{Auth}} &= z_{\text{Auth}} * \sigma_{\text{Auth}} \\
\beta_{\text{Dia}} &= \overline{\beta}_{\text{Dia}} + z_{\text{Dia}} * \sigma_{\text{Dia}} \\
\overline{\alpha}_{\text{Sent}}, \overline{\beta}_{\text{Dia}}, z_{\text{Sent}}, z_{\text{Auth}}, z_{\text{Dia}} &\sim \text{Normal}(0, 1) \\
\sigma_{\text{Sent}}, \sigma_{\text{Auth}}, \sigma_{\text{Dia}} &\sim \text{Exponencial}(1)
\end{aligned}
$$

Cópula e distância

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	1.81	0.47	1.07	2.57	1.00
β_{Dist}	-0.27	0.06	-0.37	-0.18	1.01

Classe de palavra e hierarquia de predicado

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	0.76	0.51	0.17	1.71	1
δ_{Adj}	0.48	0.18	0.19	0.78	1
δ_{Noun}	0.52	0.18	0.22	0.81	1

PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	$\beta_{\text{PoS}} * 0$	0.00	0.00	0.00	0.00	–
Adjective	$\beta_{\text{PoS}} * \delta_{\text{Adj}}$	0.35	0.26	0.06	0.86	1
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$	0.76	0.51	0.17	1.71	1

Papel temático do controlador

Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	2.73	0.60	1.88	3.76	1
δ_{Exp}	0.38	0.12	0.19	0.57	1
δ_{Agent}	0.62	0.12	0.43	0.81	1

Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	$\beta_{\text{Th}} * 0$	0.00	0.00	0.00	0.00	–
Experiencer	$\beta_{\text{Th}} * \delta_{\text{Exp}}$	1.04	0.41	0.44	1.73	1
Agent	$\beta_{\text{Th}} * (\delta_{\text{Exp}} + \delta_{\text{Agent}})$	2.73	0.60	1.88	3.76	1

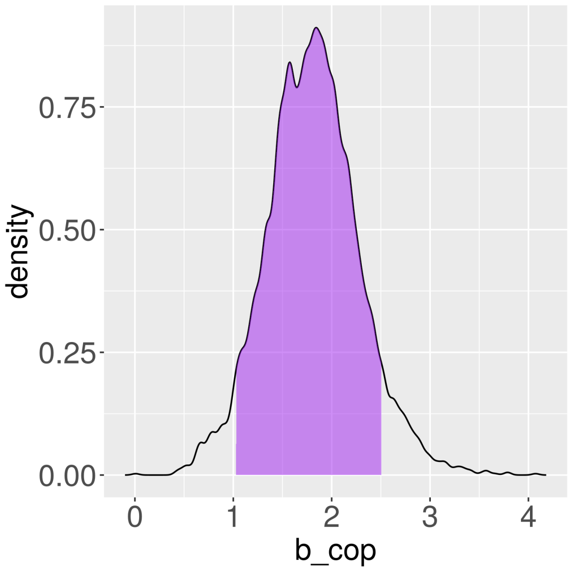


Figura 5: Distribuição posterior de β_{Cop} com o 89% do Intervalo de Maior Densidade Posterior.

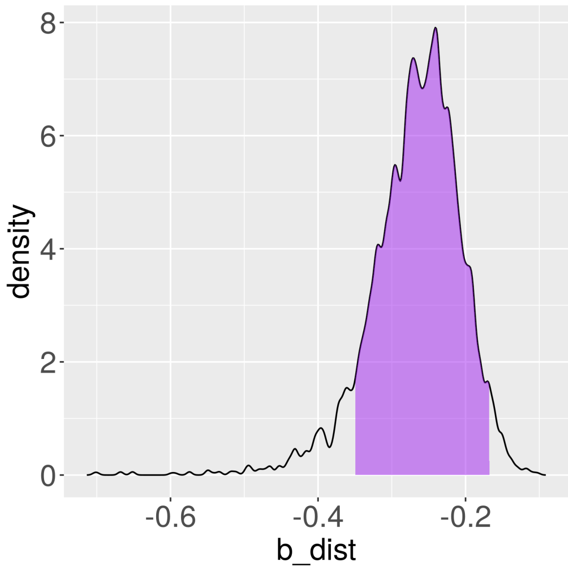


Figura 6: Distribuição posterior de β_{Dist} com o 89% do Intervalo de Maior Densidade Posterior.

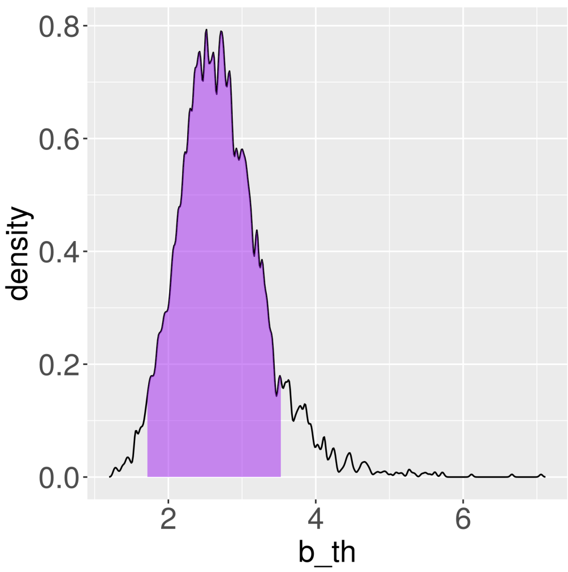


Figura 7: Distribuição posterior de β_{Th} com o 89% do Intervalo de Maior Densidade Posterior.

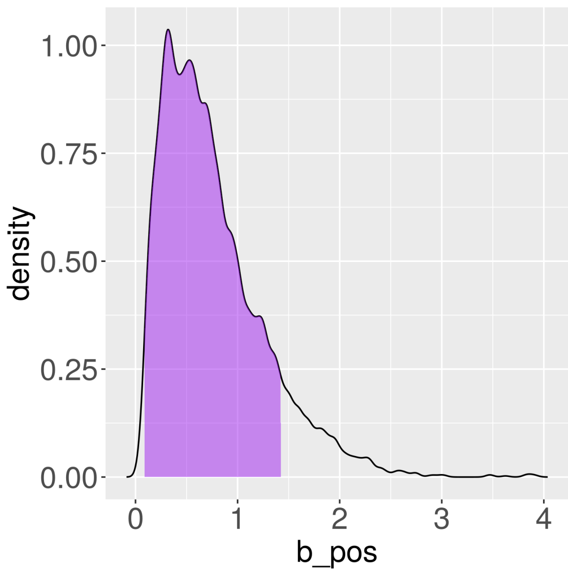


Figura 8: Distribuição posterior de β_{PoS} com o 89% do Intervalo de Maior Densidade Posterior.

Dialeto e Autoria

Effect	mean	sd	5.5%	94.5%	rhat
$\overline{\beta}_{\text{Dia}}$	-0.37	0.80	-1.64	0.93	1
σ_{Dia}	1.33	0.86	0.27	2.90	1
β_{Attic}	0.13	0.87	-1.26	1.50	1
β_{Jonic}	-1.60	1.14	-3.50	0.17	1
$\beta_{\text{Attic}} - \beta_{\text{Jonic}}$	1.73	0.94	-0.04	3.01	–

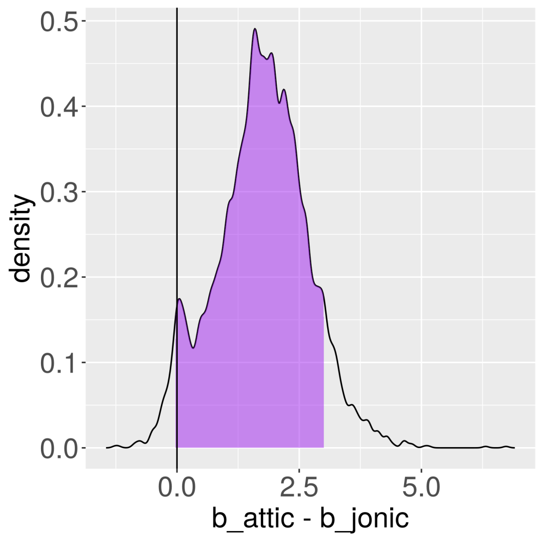


Figura 9: Distribuição posterior da diferença entre β_{Attic} e β_{Jonic} com o 89% do Intervalo de Maior Densidade Posterior.

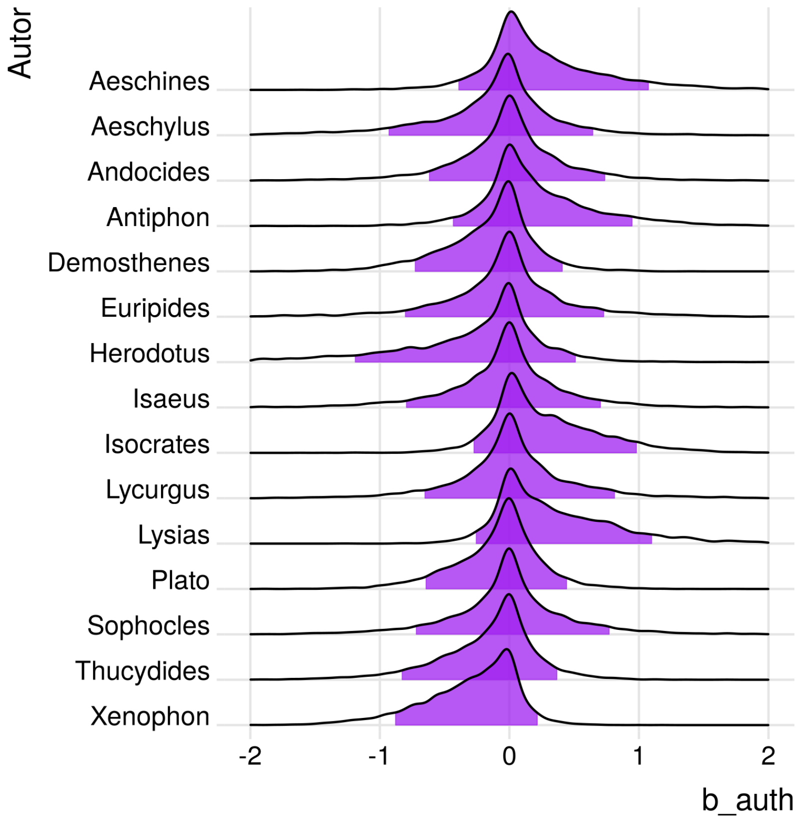


Figura 10: Distribuição posterior dos efeitos de autoria com o 89% do Intervalo de Maior Densidade Posterior.

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